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Archer **International** Schools

Class 10th, Science Chapter 5th QnA

1. Define life process.

Life processes are the basic activities that living organisms do to stay alive. These include nutrition, respiration, transportation, excretion, and reproduction. Without these, an organism cannot survive.

2. Define Nutrition.

Nutrition is the process by which organisms take in food and convert it into energy. This energy is used for growth, repair, and all life activities. It is an essential life process.

3. Difference between Autotrophic and Heterotrophic Nutrition:

Autotrophic	Heterotrophic
Organisms make their own food using sunlight.	Organisms depend on others for food.
Process used is photosynthesis.	They eat plants or animals.
Example: Plants	Example: Humans, Animals
Chlorophyll is needed.	No chlorophyll needed.

4. Define Photosynthesis (with equation):

Photosynthesis is the process by which green plants make their food using sunlight, carbon dioxide, and water. This takes place in the presence of chlorophyll.

Equation: $6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow (\text{Sunlight \& Chlorophyll}) \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$

5. Raw materials for photosynthesis:

Plants need carbon dioxide from air, water from soil, and sunlight. Chlorophyll in leaves traps sunlight and helps in food production.

6. Nutrition in Amoeba:

Amoeba captures food using finger-like extensions called pseudopodia. It forms a food vacuole, digests food inside it, and then absorbs nutrients into its body.

7. Structure of Stomata:

Stomata are tiny pores mostly found on the lower surface of leaves. Each stomata is surrounded by two kidney-shaped guard cells that control its opening and closing.

8. What are enzymes or biological catalysts?

Enzymes are proteins that speed up chemical reactions in living organisms. They help in digestion and other metabolic processes without getting used up.

9. Role of saliva in digestion:

Saliva moistens the food and makes it easier to chew and swallow. It contains the enzyme salivary amylase, which starts the digestion of starch into sugars.

10. Where do plants get raw materials for photosynthesis?

Plants absorb carbon dioxide from the air through stomata, water from the soil through roots, and sunlight from the sun. Chlorophyll in leaves helps capture sunlight.

11. What is peristalsis movement?

Peristalsis is the wave-like contraction of muscles in the food pipe. It pushes the food downward from the mouth to the stomach smoothly.

12. Role of acid in stomach:

The stomach releases hydrochloric acid which kills harmful bacteria in the food. It also creates an acidic medium for the enzyme pepsin to digest proteins.

13. Function of digestive system:

The digestive system breaks down complex food into simple substances. It helps in absorption of nutrients and removes undigested waste from the body.

14. Small intestine and absorption:

The inner lining of the small intestine has millions of villi, which are finger-like projections. These increase the surface area and help absorb nutrients into the blood.

15. The path of glucose broken down:

i. In the presence of oxygen (Aerobic respiration):

Glucose breaks into a 3-carbon compound called pyruvate. In presence of oxygen, pyruvate breaks down into carbon dioxide and water, and energy is released.

ii. In the absence of oxygen (Anaerobic respiration):

When there is no oxygen, glucose breaks down into alcohol or lactic acid. This is called fermentation in microbes.

16. Effect of haemoglobin deficiency:

Low haemoglobin means less oxygen is carried to body cells. This leads to fatigue, weakness, dizziness, and can cause anaemia.

17. Double circulation in humans & importance:

In double circulation, blood passes through the heart twice — once to the lungs and then to the body. It keeps oxygenated and deoxygenated blood separate, ensuring proper oxygen supply.

18. Transport in Xylem vs Phloem:

Xylem	Phloem
Carries water and minerals.	Carries food (glucose).
Only moves upward.	Moves in both directions.
Made of dead cells.	Made of living cells.
No energy used.	Uses energy (ATP).

19. Aerobic vs Anaerobic Respiration:

Aerobic	Anaerobic
Occurs with oxygen.	Occurs without oxygen.

Produces more energy.	Produces less energy.
End products are CO ₂ & water.	End products are alcohol or lactic acid.
Common in humans and animals.	Common in yeast, bacteria, muscles.

20. Alveoli and gas exchange:

Alveoli are tiny air sacs in lungs with thin walls and rich blood supply. Their structure provides a large surface area for quick oxygen and carbon dioxide exchange.

21. Why veins are thin walled & have valves?

Veins carry blood at low pressure, so thick walls are not needed. Valves prevent backflow and ensure blood moves toward the heart.

22. Smallest blood vessel & function:

Capillaries are the smallest blood vessels. They connect arteries and veins and help in the exchange of oxygen, nutrients, and waste between blood and body cells.

23. If arteries & veins leak:

Leaking of arteries or veins causes internal bleeding. It can reduce blood pressure, disturb oxygen supply, and may be life-threatening.

24. Difference between Arteries and Veins:

Arteries	Veins
Carry blood away from the heart.	Carry blood toward the heart.
Thick and elastic walls.	Thin walls and less elastic.
Usually oxygen-rich blood.	Usually oxygen-poor blood.
No valves.	Valves present.

25. How to measure blood pressure:

Blood pressure is measured using a sphygmomanometer. It shows two readings — systolic and diastolic — which tell how hard the heart is pumping blood.

26. Why blood mixes in amphibians & reptiles:

Amphibians and reptiles have a 3-chambered heart. This causes some mixing of oxygen-rich and oxygen-poor blood, leading to less efficient circulation.

27. Single circulation:

In single circulation, blood flows through the heart only once in one complete cycle. It is seen in fishes where blood goes from heart → gills → body → back to heart.

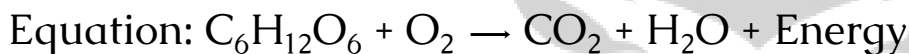
28. Composition of blood:

Blood has four main components:

- Plasma (liquid part),
 - RBCs (carry oxygen),
 - WBCs (fight germs),
 - Platelets (help in clotting).
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29. What is respiration? (with equation):

Respiration is the process of breaking down glucose to release energy for the body.



30. Glycolysis:

Glycolysis is the first step of respiration that occurs in the cytoplasm. In this step, glucose is broken into two molecules of pyruvate and a small amount of energy is released.

31. Respiratory parts of plants and animals:

Plants use stomata and lenticels for gas exchange. Animals use lungs, skin, gills, or trachea depending on their type for breathing.

32. Enzymes, Functions & Secretions:

Enzyme	Function	Secreted From
Amylase	Digests starch	Mouth, pancreas
Pepsin	Digests proteins	Stomach
Lipase	Digests fats	Pancreas
Trypsin	Continues protein digestion	Small intestine

33. Difference between Alveoli and Villi:

Alveoli	Villi
Found in lungs.	Found in small intestine.
Help in gas exchange.	Help in nutrient absorption.
Thin walls and moist surface.	Finger-like projections.
Surrounded by blood capillaries.	Contain blood vessels.

34. Composition of Blood – 5 Points:

- Blood is red fluid in our body.
- It transports oxygen, food, and hormones.
- White blood cells protect from infections.
- Platelets help in blood clotting.
- It helps maintain body temperature.

35. Which gland produces enzymes and hormones?

Pancreas produces digestive enzymes like amylase and lipase. It also makes the hormone insulin which controls blood sugar.

36. Difference between Blood and Lymph:

Blood	Lymph
Red in color due to RBCs.	Colorless fluid.
Circulates through blood vessels.	Circulates through lymph vessels.
Transports gases, food, waste.	Transports fats and fights infections.
Has more proteins and nutrients.	Has fewer proteins.

37. Define Excretion:

Excretion is the removal of harmful waste products made in the body during life processes. It is necessary to keep the body clean and healthy.

38. Where is ammonia converted to urea?

In the liver, harmful ammonia is converted into less toxic urea. This urea is then filtered out by kidneys and removed as urine.

39. Working of heart:

The heart has four chambers that pump blood in a rhythmic way. Right side pumps blood to lungs for oxygen, and left side pumps oxygen-rich blood to the rest of the body.

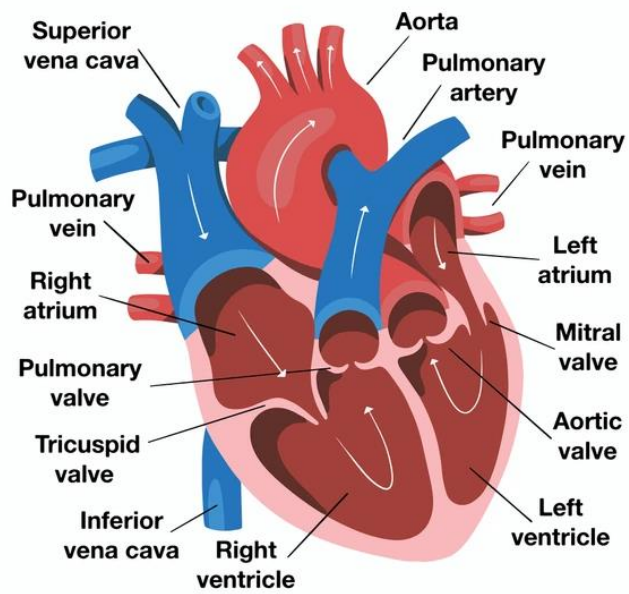
40. Heart chambers in animals:

Animal	Chambers
Fish	2 (1 atrium, 1 ventricle)
Frog	3 (2 atrium, 1 ventricle)
Snake	3 (2 atrium, 1 ventricle)
Bird & Mammals	4 (2 atrium, 2 ventricles)

41. What is Nephron?

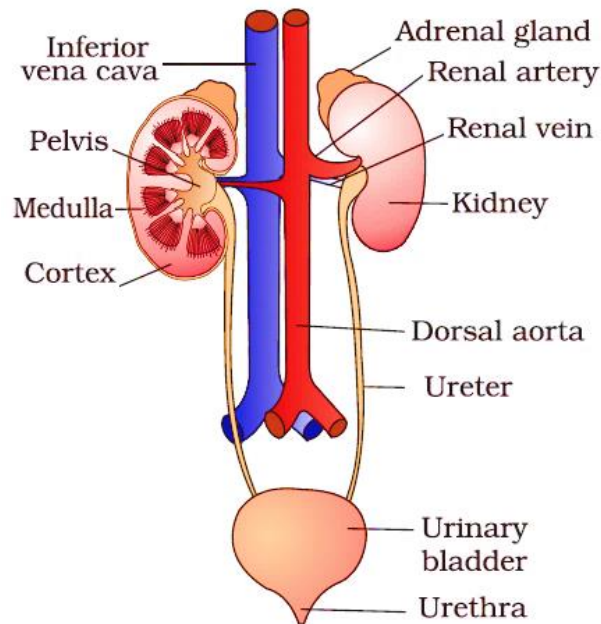
Nephron is the basic unit of the kidney. It filters the blood, removes waste like urea, and forms urine which is passed to the bladder.

42. Draw Diagram of the human heart.

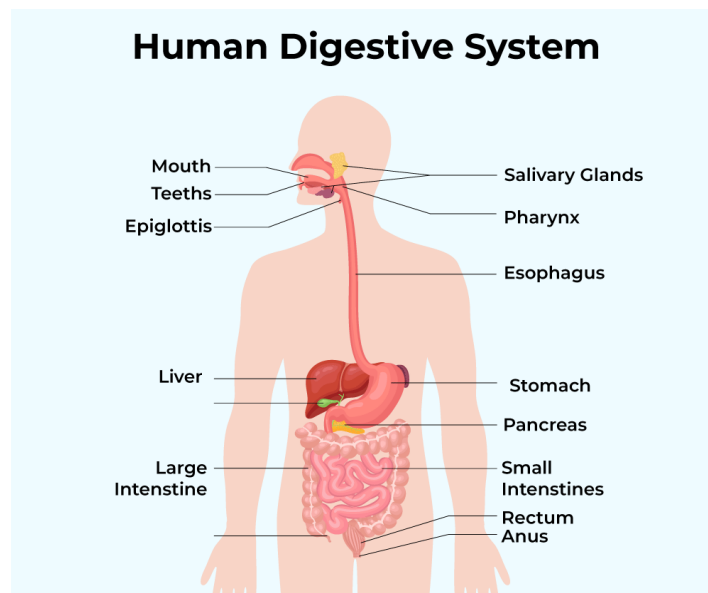


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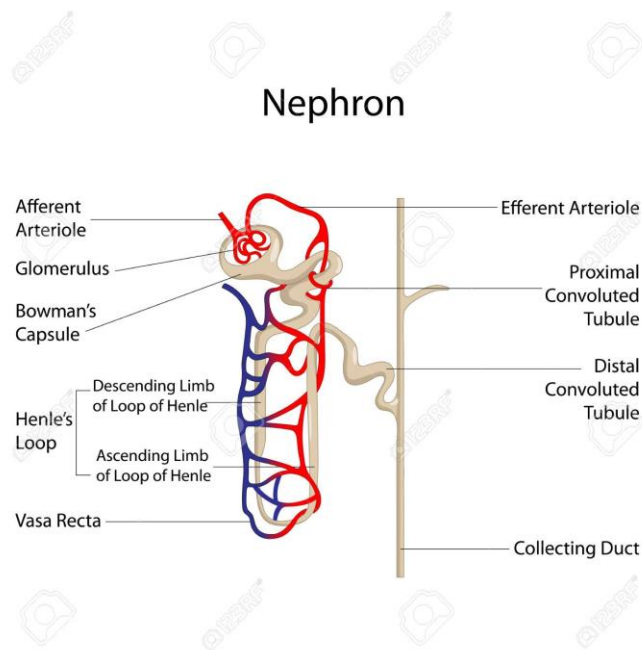
43. Draw Diagram of the human excretory system.



44. Draw Diagram of the human digestive system.

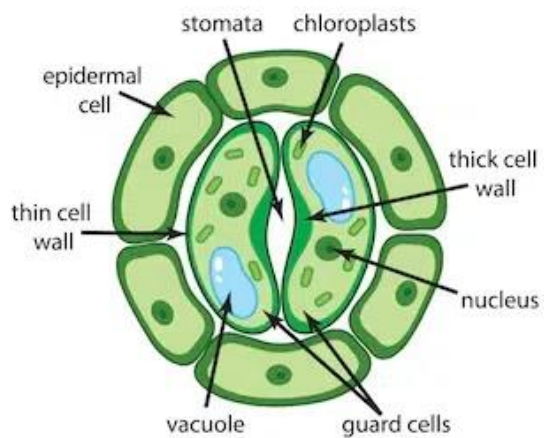


45. Draw Diagram of the nephron.

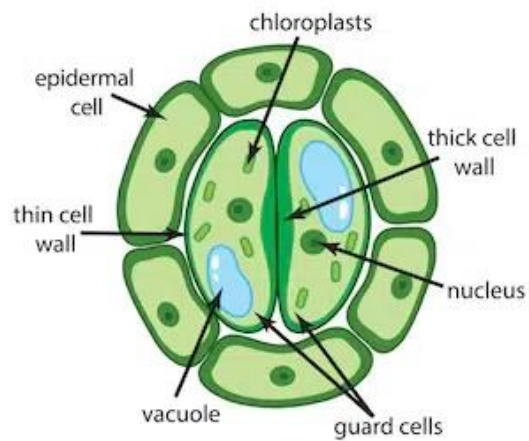


46. Draw the diagram of stomata.

Open Stomata



Closed Stomata



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